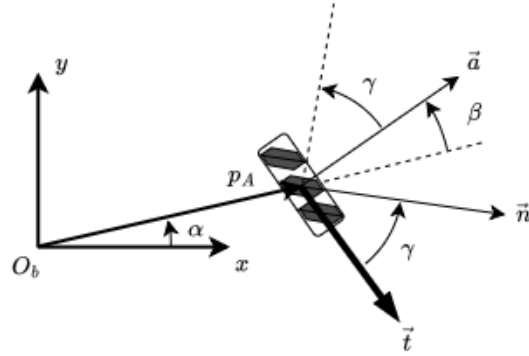


Derivation of the Omniwheel/Mecanum-Wheel Kinematic Constraint

ME 597 — Autonomous Systems — Fall 2026

An omniwheel and a mecanum wheel both have passive rollers around an actuated main wheel. For an omniwheel, the roller axes lie in the plane of the main wheel. For a mecanum wheel, the roller axes are angled, typically by $\pm\pi/4$. The rollers permit free motion perpendicular to their axes while imposing a no-slip constraint along their axes.

Consider an omniwheel or mecanum wheel of radius r , with center A , attached to a rigid body carrying the body-fixed frame $\{B\} = (x, y)$.



Let O_b denote the origin of $\{B\}$. Suppose that A lies a distance l from O_b , and let α be the signed angle from x to the wheel-position vector $\vec{P}_A$. Then

$$\vec{P}_A = \begin{bmatrix} l \cos \alpha \\ l \sin \alpha \end{bmatrix}.$$

Let the planar body-frame twist be

$$\boldsymbol{\nu}_B = \begin{bmatrix} v_x \\ v_y \\ \dot{\theta} \end{bmatrix},$$

where v_x and v_y are the translational velocity components of O_b , expressed in $\{B\}$, and $\dot{\theta}$ is the body's angular velocity. The velocity of the wheel center A is

$$\vec{v}_A = \begin{bmatrix} v_x - l\dot{\theta} \sin \alpha \\ v_y + l\dot{\theta} \cos \alpha \end{bmatrix}.$$

Let β be the signed angle from $\vec{P}_A$ to the main-wheel axle $\vec{a}$. Let $\vec{t}$ denote the main wheel's driving direction, perpendicular to $\vec{a}$, and let γ be the signed angle from $\vec{t}$ to the roller axis $\vec{n}$, with

$$\gamma \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Under these conventions, the unit vector along the roller axis is

$$\vec{n} = \begin{bmatrix} \sin \delta \\ -\cos \delta \end{bmatrix}, \quad \delta = \alpha + \beta + \gamma.$$

The passive rollers permit free motion perpendicular to their axes but impose a no-slip constraint along their axes. Consequently, the component of the wheel-center velocity along $\vec{n}$ must equal the corresponding component of the rim velocity generated by the main-wheel rotation.

Projecting $\vec{v}_A$ onto $\vec{n}$ gives

$$\begin{aligned}\vec{n}^\top \vec{v}_A &= \sin(\alpha + \beta + \gamma) \left(v_x - l\dot{\theta} \sin \alpha \right) \\ &\quad - \cos(\alpha + \beta + \gamma) \left(v_y + l\dot{\theta} \cos \alpha \right) \\ &= \begin{bmatrix} \sin(\alpha + \beta + \gamma) & -\cos(\alpha + \beta + \gamma) & -l \cos(\beta + \gamma) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \dot{\theta} \end{bmatrix}.\end{aligned}$$

Let ω be the angular velocity of the main wheel. The tangential rim velocity generated by the wheel rotation is $r\omega\vec{t}$. Because the angle between $\vec{t}$ and $\vec{n}$ is γ , its component along the roller axis is

$$\vec{n}^\top (r\omega\vec{t}) = r\omega \cos \gamma.$$

Equating the two velocity components yields the wheel constraint

$$\begin{bmatrix} \sin(\alpha + \beta + \gamma) & -\cos(\alpha + \beta + \gamma) & -l \cos(\beta + \gamma) \end{bmatrix} \boldsymbol{\nu}_B = r\omega \cos \gamma.$$

If the vehicle configuration is expressed in the inertial frame as

$$\boldsymbol{\xi}_I = \begin{bmatrix} X_e \\ Y_e \\ \theta \end{bmatrix},$$

define the inertial-to-body twist transformation as

$$\boldsymbol{\nu}_B = \mathcal{R}(\theta)\dot{\boldsymbol{\xi}}_I, \quad \mathcal{R}(\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The constraint written in terms of the inertial-frame configuration rate is therefore

$$\boxed{\begin{bmatrix} \sin(\alpha + \beta + \gamma) & -\cos(\alpha + \beta + \gamma) & -l \cos(\beta + \gamma) \end{bmatrix} \mathcal{R}(\theta)\dot{\boldsymbol{\xi}}_I = r\omega \cos \gamma}.$$

The signs in this expression depend on the adopted conventions for positive wheel rotation, the orientation of $\vec{t}$, and the signed angles α , β , and γ . These conventions should therefore be fixed consistently when constructing the multiwheel kinematic model.